

Fast matrix multiplication using CP decomposition

Computational laboratory project report

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Abstract

Bla bla bla

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1 Notation and tensor preliminaries

A tensor is a multidimensional array, and in this work we deal only with 3-dimensional (or 3-way) tensors $\mathcal{X} \in \mathbb{R}^{m \times n \times p}$.

Short
recap of
tensor
stuff

2 Low rank tensor decomposition

3 Fast matrix multiplication

Matrix multiplication is a bilinear operation. We indicate with $\langle m, n, p \rangle$ the bilinear form that represents the multiplication between a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ with $\mathbf{B} \in \mathbb{R}^{n \times p}$. Fixing the natural ordering on the entries of $\mathbf{A}$ and $\mathbf{B}$ and the inverse ordering on $\mathbf{C} = \mathbf{A}\mathbf{B}$, we get a representation of $\mathcal{X} \in \mathbb{R}^{mn \times np \times mp}$ of the matrix multiplication $\langle m, n, p \rangle$ such that

$$(1) \quad \text{vec}(\mathbf{C}^\top) = \mathcal{X} \times_1 \text{vec}(\mathbf{A})^\top \times_2 \text{vec}(\mathbf{B})^\top.$$

Note that we fix the linear order on A and B , while we fix the reverse ordering on C by applying the transpose operation.

why

If we are given a CP decomposition of rank r of the tensor $\mathcal{X} = \llbracket U, V, W \rrbracket$, we can rewrite Equation 1 into

Tensor
definition

$$(2) \quad \text{vec}(\mathbf{C}^\top) = \sum_{l=1}^r (\underbrace{\mathbf{u}_l^\top \text{vec}(\mathbf{A})}_{s_l}) \cdot (\underbrace{\mathbf{v}_l^\top \text{vec}(\mathbf{B})}_{t_l}) \mathbf{w}_l = \sum_{l=1}^r (\underbrace{s_l \cdot t_l}_{m_l}) w_l,$$

where $\mathbf{u}_1, \mathbf{v}_1, \mathbf{w}_1$ denote the columns of $\mathbf{U}$, $\mathbf{V}$ and $\mathbf{W}$. This reduces the number of active multiplications to the rank r of $\mathcal{X}$.

When $m, n, p > 2$ we can always partition them in blocks as:

$$\begin{matrix} \overbrace{[n/2]} & \overbrace{[n/2]} \\ \lfloor m/2 \rfloor \{ \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, & \lfloor n/2 \rfloor \{ \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}, \\ \overbrace{[m/2]} & \overbrace{[n/2]} \end{matrix}$$

where $A_{11} \in \mathbb{R}^{\lfloor \frac{m}{2} \rfloor \times \lfloor \frac{n}{2} \rfloor}$, $B_{11} \in \mathbb{R}^{\lfloor \frac{n}{2} \rfloor \times \lfloor \frac{p}{2} \rfloor}$, $A_{22} \in \mathbb{R}^{\lceil \frac{m}{2} \rceil \times \lceil \frac{n}{2} \rceil}$, and $B_{22} \in \mathbb{R}^{\lceil \frac{n}{2} \rceil \times \lceil \frac{p}{2} \rceil}$.

In particular, we can view the product AB as the product between 2×2 block matrices

$$(3) \quad \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \cdot \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix}.$$

The naive algorithm proceeds by combining a set of eight matrix multiplications with four additions:

$$\begin{aligned} M_1 &= A_{11} \cdot B_{11} & M_2 &= A_{12} \cdot B_{21} & M_3 &= A_{11} \cdot B_{12} \\ M_4 &= A_{12} \cdot B_{22} & M_5 &= A_{21} \cdot B_{11} & M_6 &= A_{22} \cdot B_{21} \\ M_7 &= A_{21} \cdot B_{12} & M_8 &= A_{22} \cdot B_{22} \end{aligned}$$

$$\begin{aligned} C_{11} &= M_1 + M_2 & C_{12} &= M_3 + M_4 \\ C_{21} &= M_5 + M_6 & C_{22} &= M_7 + M_8 \end{aligned}$$

For example, when $m = n = p = 2$, we get the tensor with frontal slices

$$\mathbf{X}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{X}_2 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{X}_3 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad \mathbf{X}_4 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

This yields, for example,

$$X_3 \times_1 \text{vec}(A)^\top \times_2 \text{vec}(B)^\top = \begin{bmatrix} a_{11} & a_{21} & a_{12} & a_{22} \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \\ b_{12} \\ b_{22} \end{bmatrix} = a_{21}b_{11} + a_{22}b_{21} = c_{21}.$$

Note that the formula in Equation 2 in the context of block matrix multiplication in Equation 3 yields to

$$\text{vec}(A) = \begin{bmatrix} A_{11} \\ A_{21} \\ A_{12} \\ A_{22} \end{bmatrix}, \quad \text{vec}(B) = \begin{bmatrix} B_{11} \\ B_{21} \\ B_{12} \\ B_{22} \end{bmatrix}, \quad \text{vec}(C^\top) = \begin{bmatrix} C_{11} \\ C_{12} \\ C_{21} \\ C_{22} \end{bmatrix}.$$

Also the factors s_l and t_l in Equation 2 become

$$s_l = u_l^\top \text{vec}(A) = \sum_{i=1}^4 U_{li} \text{vec}(A)_i, \quad t_l = V_l^\top \text{vec}(B) = \sum_{i=1}^4 V_{li} \text{vec}(B)_i$$

References